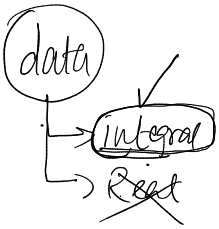
Floating Point data (Real Data) follows IEEE-754 Std.

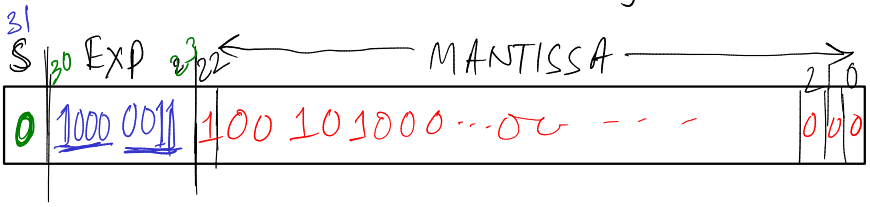
the Std says Every real Data has 3 Parts.

→ Sign, Exponent, Mantissa

	Sign	Exp	Mantissa	Bias value
float (32 bits)	1 bit	8 bits	23 bits	127 (0x7F)
double (64 bits)	1 bit	11 bits	52 bits	1023 (0x3FF)



float var = 25.25 ; // var in memory occupies 4 bytes



0x7F + 4 = 0x83

d.ddd e±n (1-9)

25 => 11001

I) $(25.25)_{10} \Rightarrow (11001.01)_2$

0.25 x 2 = 0.5 | 0

0.5 x 2 = 1.0 | 1

II => Use Normalized form | Exponential form

± 1.bbbbbb e±n

Sign, mantissa, Exp

$(11001.01)_2 \Rightarrow +1.100101\ e+4$

M, E, S

-ve integers/chars are stored in 2's compliment form

int x = 1, y = -1; x = -50; y = 50

-ve Real Data Not stored in 2's Comp form

→ that means float x = 123.45 and float y = -123.45 will have Exact SAME BIN EQ. except the Sign Bit

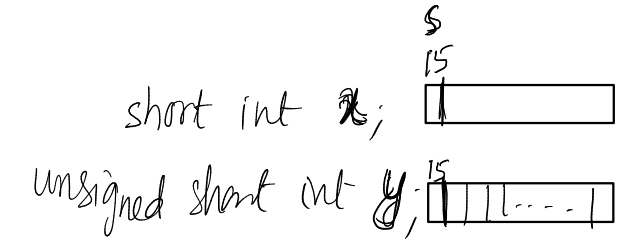
float V = -50.125

Step I ⇒ $(-50.125)_{10} \Rightarrow -(110010.001)_2$

$0.125 \times 2 = 0.25 \mid 0$
$0.25 \times 2 = 0.5 \mid 0$
$0.5 \times 2 = 1.0 \mid 1$

II ⇒ $-1.10010001e+5$

S	E	m
1	10000100	10010001000000



float V = 0.0625

$(0.0625)_{10} = (0.0001)_2$

$+1.0e-4$

$0.0625 \times 2 = 0.125 \mid 0$
$0.125 \times 2 = 0.25 \mid 0$
$0.25 \times 2 = 0.5 \mid 0$
$0.5 \times 2 = 1.0 \mid 1$

0 | 0111 1011 | 000 ... 0

0111 1111 = 127

0111 1011

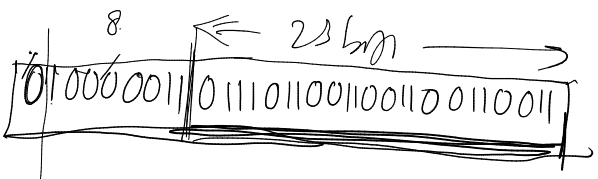
$(23.4)_{10} = (10111.0110)_2$

$\Downarrow$

$+1.0111\overline{0110}\ e+4$

$0.4 \times 2 = 0.8$	0
$0.8 \times 2 = 1.6$	1
$0.6 \times 2 = 1.2$	1
$0.2 \times 2 = 0.4$	0

$0.4 \times 2 = 0.8$	0
$0.8 \times 2 = 1.6$	1
$0.6 \times 2 = 1.2$	1
:	0
:	:

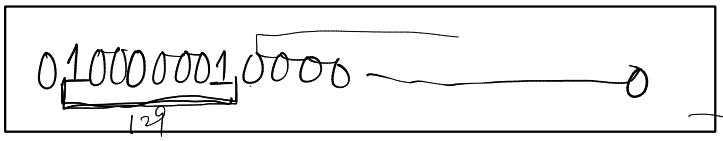


1 - Why unsigned float not allowed

2 - Why some floating point data are slightly (negligibly) modified ?

3. Range of float ? How 32 bits float having range much higher than 2^{32}.

In a float variable the bin.. eq. found as



If used printf("%f", var);

What will be the output ?

$S\left(\frac{E}{127}\right)m$

2

$1.0\ e+2$

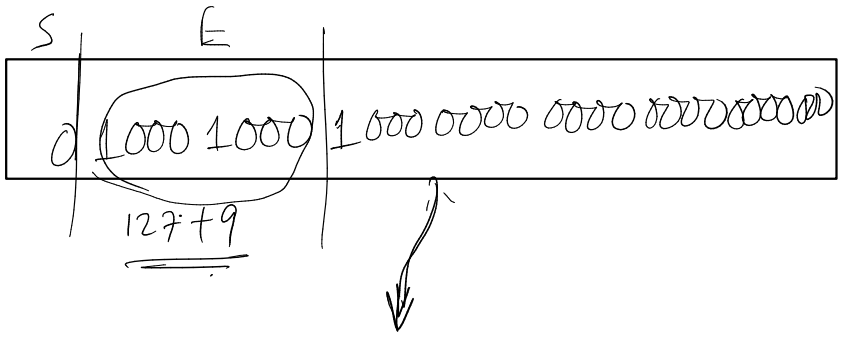
$\Downarrow$

$(100.0000\ldots)_2 = (4.0)_2$

In a float var;

≡

the bin eq. appears as



What's the Decimal eq.?

Pf("%f", var);

?

1.m e+9

1.1e+9

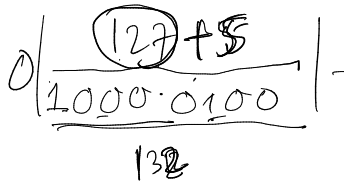
$(1100000000.0)_2$

2^9+2^8

512+256

768.0

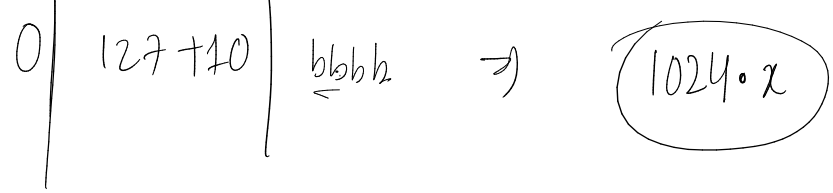
Consider in a float variable the data in bits like



1.0 e+5

$(100000.0)_2$

What's the value? 32.0

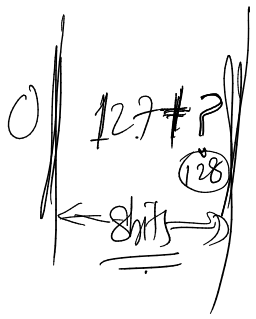


0 | 127 #20 | 0 ⇒ ? 10^6

1 Mn.

0 | 127 +30 |

⇒ 10^9



⇒ ? $2^{128} = 2^{64} \times 2^{64}$

$= 1.8 \times 10^{19} \times 1.8 \times 10^{19}$

$= 3.4 \times 10^{38}$

---

int x = 20; int r1;

float y = 20.0; float r2;

x ⇒ S 00000000 00000000000000 00010100

y ⇒ S 10000001 10100 ~ ~

r1 = x+y; r2 = x+y;

40.0 40.0

Pf("r1 = %d \n", r1); Pf("r2 = %f", r2);

40 40.0

10100.0 → 1.01 e+4 ⇒

01000001 010 ~

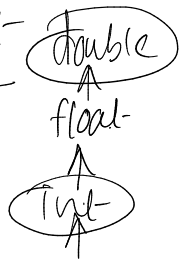
area = 3.14 * rad * rad;

Typecast (appearing ~~casting~~ as a diff type)
temporary type conversion

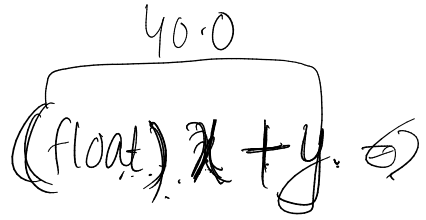
Implicit
(default followed by compiler)

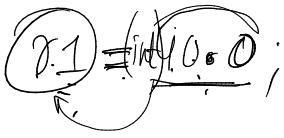
Explicit
(programmers choice)

Upper type cast

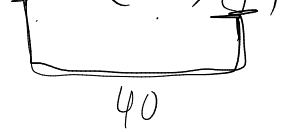


r1 =
r2 =





r2 = 40.0



---

next class Arithmetic, logical, relational